task2.py | task3.py | task4.py | task5.py

task1.py > ...

```
1   start = int(input("Enter starting weekday (0=Sun, 1
2   days = int(input("Enter number of days in the month
3   if start < 0 or start > 6 or days < 28 or days > 31
4       print("Invalid input")
5   else:
6       print("Sun Mon Tue Wed Thu Fri Sat")
7
8       for i in range(start):
9           print("    ", end="")
10      day = 1
11      count = start
12      while day <= days:
13          for j in range(7):
14              if day <= days and count >= start:
15                  print("%3d " % day, end="")
16                  day += 1
17              else:
18                  print("    ", end="")
19              count += 1
20          print()
21
```

Problems | Output | Debug Console | Terminal | Ports

```
madhurima@madhurimas-Air ABL % /usr/local/bin/python3 /User
Enter starting weekday (0=Sun, 1=Mon, ..., 6=Sat): 3
Enter number of days in the month: 30
Sun Mon Tue Wed Thu Fri Sat
                1   2   3   4   5   6   7
  8   9  10  11  12  13  14
 15  16  17  18  19  20  21
 22  23  24  25  26  27  28
 29  30
madhurima@madhurimas-Air ABL %
```

task2.py task3.py task4.py task5.py

task2.py > ...

```python
rows = 4
cols = 5

reserved = [(1, 2), (3, 4)]
seat_no = 1

for i in range(1, rows + 1):
    for j in range(1, cols + 1):
        if (i, j) in reserved:
            print("RES", end="\t")
        elif (i + j) % 2 == 0:
            print("X", end="\t")
        else:
            print(seat_no, end="\t")
            seat_no += 1
    print()

```

Problems Output Debug Console Terminal Ports

```
madhurima@madhurimas-Air ABL % /usr/local/bin/python3 /User
Enter number of days in the month: 30
Sun Mon Tue Wed Thu Fri Sat
                1   2   3   4   5   6   7
  8   9  10  11  12  13  14
 15  16  17  18  19  20  21
 22  23  24  25  26  27  28
 29  30
madhurima@madhurimas-Air ABL % /usr/local/bin/python3 /User
X       RES     X       1       X
2       X       3       X       4
X       5       X       RES     X
6       X       7       X       8
madhurima@madhurimas-Air ABL %
```

task2.py | task3.py | task4.py | task5.py

task3.py > ...

```python
import math
points = [(0, 0), (3, 4), (6, 8), (12, 8), (12, 16)
longest = 0
segment = 0
for i in range(len(points) - 1):
    x1, y1 = points[i]
    x2, y2 = points[i + 1]
    distance = math.sqrt((x2 - x1)**2 + (y2 - y1)**
    print("Segment", i + 1, "Distance:", distance)
    if distance > longest:
        longest = distance
        segment = i + 1
print("Longest Route Segment:", segment)
print("Longest Distance:", longest)

```

Problems | Output | Debug Console | Terminal | Ports

```
madhurima@madhurimas-Air ABL % /usr/local/bin/python3 /User
madhurima@madhurimas-Air ABL % /usr/local/bin/python3 /User
X       RES     X       1       X
2       X       3       X       4
X       5       X       RES     X
6       X       7       X       8
madhurima@madhurimas-Air ABL % /usr/local/bin/python3 /User
Segment 1 Distance: 5.0
Segment 2 Distance: 5.0
Segment 3 Distance: 6.0
Segment 4 Distance: 8.0
Longest Route Segment: 4
Longest Distance: 8.0
madhurima@madhurimas-Air ABL %
```

task2.py | task3.py | task4.py | task5.py

task4.py > students

```python
students = ["Arun", "Bala", "Chitra"]
exam1 = [60, 35, 80]
exam2 = [70, 30, 85]
exam3 = [80, 25, 90]
for i in range(3):
    print("\nStudent:", students[i])
    avg = (exam1[i] + exam2[i] + exam3[i]) / 3
    print("Average Marks:", avg)
    change = exam3[i] - exam1[i]
    print("Change in Marks:", change)
    if change > 0:
        print("Trend: Improved")
    elif change < 0:
        print("Trend: Declined")
    else:
        print("Trend: No Change")
    if avg >= 80:
        print("Category: Excellent")
    elif avg >= 60:
        print("Category: Good")
    elif avg >= 40:
        print("Category: Average")
    else:
```

Problems | Output | Debug Console | Terminal | Ports

```
madhurima@madhurimas-Air ABL % /usr/local/bin/python3 /User
Average Marks: 30.0
Change in Marks: -10
Trend: Declined
Category: Needs Improvement
Weak Performance: Yes

Student: Chitra
Average Marks: 85.0
Change in Marks: 10
Trend: Improved
Category: Excellent
Weak Performance: No
madhurima@madhurimas-Air ABL %
```

task2.py | task3.py | task4.py | task5.py

task5.py > teams

```python
teams = [("Team A", [101, 102]),
    ("Team B", [103, 104, 105]),
    ("Team C", [106, 107])]
scores = [[80, 90, 85],
    [90, 85, 80],
    [80, 90, 85]]
min_size = 2
max_size = 3
totals = []
for i in range(len(teams)):
    name = teams[i][0]
    members = teams[i][1]
    print(name)
    if len(members) < min_size or len(members) > ma
        print("Invalid Team Size")
        total = -1
    else:
        print("Team Size:", len(members))

        total = 0
        for j in range(len(scores[i])):
            total += scores[i][j]
        print("Total Score:", total)
```

Problems | Output | Debug Console | Terminal | Ports

```
madhurima@madhurimas-Air ABL % /usr/local/bin/python3 /User
Total Score: 255
Team B
Team Size: 3
Total Score: 255
Team C
Team Size: 2
Total Score: 255

Final Rankings
Rank 1 : Team A - 255
Rank 1 : Team B - 255
Rank 1 : Team C - 255
madhurima@madhurimas-Air ABL %
```

task2.py | task3.py | task4.py | task5.py

task6.py > ...

```python
email = input("Enter student email: ")
reg_id = input("Enter registration ID: ")
if "@" not in email:
    print("Email Error: Missing @ symbol")
elif email.count("@") != 1:
    print("Email Error: Multiple @ symbols")
elif "." not in email:
    print("Email Error: Missing dot")
elif " " in email:
    print("Email Error: Contains spaces")
elif email.startswith("@") or email.endswith("@"):
    print("Email Error: Missing email component")
elif email.startswith(".") or email.endswith("."):
    print("Email Error: Invalid dot position")
else:
    print("Email & Registration ID: Valid")
```

Problems Output Debug Console **Terminal** Ports

```
madhurima@madhurimas-Air ABL % /usr/local/bin/python3 /User
Team C
Team Size: 2
Total Score: 255

Final Rankings
Rank 1 : Team A - 255
Rank 1 : Team B - 255
Rank 1 : Team C - 255
madhurima@madhurimas-Air ABL % /usr/local/bin/python3 /User
Enter student email: your@gmail.com
Enter registration ID: ST4321
Email & Registration ID: Valid
madhurima@madhurimas-Air ABL %
```

task2.py | task3.py | task4.py | task5.py

task7.py > registrations

```python
registrations = [(101, "Python"),(101, "Maths"),(10
valid_courses = ("Python", "Maths", "Java")
invalid_pairs = (("Python", "Java"),)
students = [101, 102, 103]
for student in students:
    courses = []
    for reg in registrations:
        if reg[0] == student:
            courses.append(reg[1])
    print("\nStudent:", student)
    print("Courses:", courses)
    duplicate = False
    for i in range(len(courses)):
        for j in range(i + 1, len(courses)):
            if courses[i] == courses[j]:
                duplicate = True
    if duplicate:
        print("Duplicate Registration Found")
    invalid = False
    for course in courses:
        if course not in valid_courses:
            invalid = True
    if invalid:
```

Problems | Output | Debug Console | Terminal | Ports

```
madhurima@madhurimas-Air ABL % /usr/local/bin/python3 /User

Student: 101
Courses: ['Python', 'Maths']
Registration Valid

Student: 102
Courses: ['Python', 'Python']
Duplicate Registration Found

Student: 103
Courses: ['Python', 'Java']
Invalid Course Combination
madhurima@madhurimas-Air ABL %
```

task2.py | task3.py | task4.py | task5.py

task8.py > students

```python
students = ["Arun", "Bala", "Chitra", "Deepak",
            "Esha", "Farhan"]
scores = [80, 60, 90, 70, 85, 65]
group_size = 2
groups = [[], [], []]
totals = [0, 0, 0]
for i in range(len(scores)):
    for j in range(i + 1, len(scores)):
        if scores[j] > scores[i]:
            scores[i], scores[j] = scores[j], score
            students[i], students[j] = students[j],
for i in range(len(students)):
    min_group = 0
    for j in range(1, len(groups)):
        if totals[j] < totals[min_group]:
            min_group = j
    groups[min_group].append(students[i])
    totals[min_group] += scores[i]
print("Student Groups:")
for i in range(len(groups)):
    print("Group", i + 1, ":", groups[i])
    print("Group Total Score:", totals[i])
difference = max(totals) - min(totals)
```

Problems | Output | Debug Console | Terminal | Ports

```
madhurima@madhurimas-Air ABL % /usr/local/bin/python3 /User
Student: 103
Courses: ['Python', 'Java']
Invalid Course Combination
madhurima@madhurimas-Air ABL % /usr/local/bin/python3 /User
Student Groups:
Group 1 : ['Chitra', 'Bala']
Group Total Score: 150
Group 2 : ['Esha', 'Farhan']
Group Total Score: 150
Group 3 : ['Arun', 'Deepak']
Group Total Score: 150
Difference between group totals: 0
madhurima@madhurimas-Air ABL %
```

task2.py | task3.py | task4.py | task5.py

task9.py > answer_key

```python
answer_key = ["A", "B", "C", "D", "A"]
students = ["Arun", "Bala", "Chitra"]
answers = [["A", "B", "C", "D", "A"],
    ["A", "C", "", "D", "B"],
    ["B", "B", "C", "", "A"]]
correct_mark = 2
negative_mark = 0.5
for i in range(len(students)):
    correct = 0
    incorrect = 0
    unanswered = 0
    for j in range(len(answer_key)):
        if answers[i][j] == "":
            unanswered += 1
        elif answers[i][j] == answer_key[j]:
            correct += 1
        else:
            incorrect += 1
    score = (correct * correct_mark) - (incorrect *
    print("Student:", students[i])
    print("Correct Answers:", correct)
    print("Incorrect Answers:", incorrect)
    print("Unanswered Questions:", unanswered)
```

Problems | Output | Debug Console | Terminal | Ports

```
madhurima@madhurimas-Air ABL % /usr/local/bin/python3 /User
Unanswered Questions: 0
Final Score: 10.0
Student: Bala
Correct Answers: 2
Incorrect Answers: 2
Unanswered Questions: 1
Final Score: 3.0
Student: Chitra
Correct Answers: 3
Incorrect Answers: 1
Unanswered Questions: 1
Final Score: 5.5
madhurima@madhurimas-Air ABL %
```

Navigate to Command
Scroll to Previous Command (⌘
Scroll to Next Command (⌘↓)

task2.py | task3.py | task4.py | task5.py

task10.py > students

```python
students = ["Arun", "Bala", "Chitra", "Deepak"]
marks = [75, 35, 55, 40]
attendance = [85, 60, 75, 65]
lab_scores = [80, 30, 45, 35]
high_risk = 0
medium_risk = 0
low_risk = 0
for i in range(len(students)):
    risk = 0
    print("\nStudent:", students[i])
    if marks[i] < 40:
        print("Low Marks")
        risk += 1
    if attendance[i] < 75:
        print("Low Attendance")
        risk += 1
    if lab_scores[i] < 40:
        print("Low Lab Score")
        risk += 1
    if risk >= 2:
        category = "High Risk"
        high_risk += 1
    elif risk == 1:
```

Problems Output Debug Console **Terminal** Ports

```
madhurima@madhurimas-Air ABL % /usr/local/bin/python3 /User
Category: Low Risk

Student: Deepak
Low Attendance
Low Lab Score
Risk Factors: 2
Category: High Risk

ACADEMIC RISK SUMMARY
High Risk Students: 2
Medium Risk Students: 0
Low Risk Students: 2
madhurima@madhurimas-Air ABL %
```